

Code:

```
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J GradeCalculator.java X
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1 import javax.swing.*;
2 import java.text.DecimalFormat;
3 import java.util.ArrayList;
4
5 public class GradeCalculator {
6     private static final ArrayList<Double> grades = new ArrayList<>();
7     private static final DecimalFormat df = new DecimalFormat("0.00");
8
9     /**
10      * Program entry point.
11      * Shows welcome message, Loops through menu until exit, then shows goodbye message.
12      */
13     public static void main(String[] args) {
14         showWelcomeMessage();
15         int choice;
16         do {
17             choice = showMenu();
18             processChoice(choice);
19         } while (choice != 4);
20         showGoodbyeMessage();
21     }
22
23     /**
24      * Shows welcome message to the user at program start.
25      */
26     private static void showWelcomeMessage() {
27         JOptionPane.showMessageDialog(null,
28             "Welcome to the Grade Calculator System!\n\n" +
29             "This program will help you track your grades\n" +
30             "and calculate your current average.",
31             "Grade Calculator",
```

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31         "Grade Calculator",
32         JOptionPane.INFORMATION_MESSAGE);
33     }
34
35     /**
36      * Displays the main menu and returns the user's choice.
37      *
38      * @return menu option chosen by the user (1-4), or -1 if invalid
39      */
40     private static int showMenu() {
41         String menu = ""
42             + "Grade Calculator\n\n"
43             + "1. Add Grade\n"
44             + "2. View Average\n"
45             + "3. View Letter Grade\n"
46             + "4. Exit\n\n";
47
48         String input = JOptionPane.showInputDialog(null, menu, "Menu", JOptionPane.QUESTION_MESSAGE);
49
50         if (input == null) {
51             return 4; // User canceled
52         }
53         try {
54             return Integer.parseInt(input);
55         } catch (NumberFormatException e) {
56             return -1; // Invalid input
57         }
58     }
59
60 }
61
```

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62 /**
63  * Processes the user's menu choice and calls the appropriate method.
64  *
65  * @param choice menu option selected by the user
66  */
67 private static void processChoice(int choice) {
68     switch (choice) {
69         case 1 -> addGrade();
70         case 2 -> viewAverage();
71         case 3 -> viewLetterGrade();
72         case 4 -> {} // goodbye handled after loop
73         default -> JOptionPane.showMessageDialog(null, "Invalid choice. Try again.");
74     }
75 }
76
77 /**
78  * Prompts the user to enter a grade (0-100) and adds it to the List if valid.
79  */
80 private static void addGrade() {
81     String input = JOptionPane.showInputDialog(null, "Enter grade (0-100):");
82     if (input == null) return;
83     try {
84         double grade = Double.parseDouble(input);
85         if (grade < 0 || grade > 100) {
86             JOptionPane.showMessageDialog(null, "Grade must be between 0 and 100.");
87         } else {
88             grades.add(grade);
89             JOptionPane.showMessageDialog(null, "Grade added: " + df.format(grade));
90         }
91     } catch (NumberFormatException e) {
92         JOptionPane.showMessageDialog(null, "Invalid number. Please enter a valid grade.");
93     }
94 }
95
96 /**
97  * Calculates and displays the average grade with snarky/funny feedback.
98  */
99 private static void viewAverage() {
100     if (grades.isEmpty()) {
101         JOptionPane.showMessageDialog(null, "No grades entered yet.", "Average", JOptionPane.INFORMATION_MESSAGE);
102         return;
103     }
104     double avg = calculateAverage();
105     String message = "Average of " + grades.size() + " grade(s): " + df.format(avg) + "%";
106
107     // Funny/snarky feedback
108     String feedback;
109     if (avg >= 90) {
110         feedback = "\nBruh you're built different. Straight A Vibes!";
111     } else if (avg >= 80) {
112         feedback = "\nSolid work, you're lowkey cruising.";
113     } else if (avg >= 70) {
114         feedback = "\nMid-tier grind could be worse, could be better.";
115     } else if (avg >= 60) {
116         feedback = "\nBarely surviving. You're literally speedrunning stress.";
117     } else {
118         feedback = "\nOof that grade flopped harder than a failed TikTok.";
119     }
120
121     JOptionPane.showMessageDialog(null, message + feedback, "Average", JOptionPane.INFORMATION_MESSAGE);
122 }
123
```

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93 }
94 }
95
96 /**
97  * Calculates and displays the average grade with snarky/funny feedback.
98  */
99 private static void viewAverage() {
100     if (grades.isEmpty()) {
101         JOptionPane.showMessageDialog(null, "No grades entered yet.", "Average", JOptionPane.INFORMATION_MESSAGE);
102         return;
103     }
104     double avg = calculateAverage();
105     String message = "Average of " + grades.size() + " grade(s): " + df.format(avg) + "%";
106
107     // Funny/snarky feedback
108     String feedback;
109     if (avg >= 90) {
110         feedback = "\nBruh you're built different. Straight A Vibes!";
111     } else if (avg >= 80) {
112         feedback = "\nSolid work, you're lowkey cruising.";
113     } else if (avg >= 70) {
114         feedback = "\nMid-tier grind could be worse, could be better.";
115     } else if (avg >= 60) {
116         feedback = "\nBarely surviving. You're literally speedrunning stress.";
117     } else {
118         feedback = "\nOof that grade flopped harder than a failed TikTok.";
119     }
120
121     JOptionPane.showMessageDialog(null, message + feedback, "Average", JOptionPane.INFORMATION_MESSAGE);
122 }
123
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124 /**
125  * Calculates and displays the Letter grade with snarky/funny feedback.
126  */
127 private static void viewLetterGrade() {
128     if (grades.isEmpty()) {
129         JOptionPane.showMessageDialog(null, "No grades entered yet.", "Letter Grade", JOptionPane.INFORMATION_MESSAGE);
130         return;
131     }
132     double avg = calculateAverage();
133     String letter = getLetterGrade(avg);
134     String message = "Average: " + df.format(avg) + "\nLetter grade: " + letter;
135
136     // Funny/snarky feedback
137     String feedback;
138     switch (letter) {
139         case "A", "A-" -> feedback = "\nTop of the food chain, go off queen";
140         case "B+", "B", "B-" -> feedback = "\nRespectable not main character energy, but close.";
141         case "C+", "C", "C-" -> feedback = "\nBare minimum detected. Might need a glow-up.";
142         case "D+", "D", "D-" -> feedback = "\nDanger zone! This GPA is fighting for its life.";
143         default -> feedback = "\nRip pack it up, chief. Summer school arc unlocked.";
144     }
145
146     JOptionPane.showMessageDialog(null, message + feedback, "Letter Grade", JOptionPane.INFORMATION_MESSAGE);
147 }
148
149 /**
150  * Displays final average and Letter grade when the program ends, with a vibe-check message.
151  */
152 private static void showGoodbyeMessage() {
153     String message;
154     if (grades.isEmpty()) {
```

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155     message = "Goodbye! You didn't enter any grades.\nLiterally no data, no drama.";
156 } else {
157     double avg = calculateAverage();
158     String letter = getLetterGrade(avg);
159     message = "Goodbye!\nFinal average: " + df.format(avg) +
160         "\nFinal letter grade: " + letter;
161
162     if (avg >= 90) {
163         message += "\nYou're built like a study machine.";
164     } else if (avg >= 75) {
165         message += "\nNot bad keep cruising, you'll be fine.";
166     } else if (avg >= 60) {
167         message += "\nBro clutch up next time.";
168     } else {
169         message += "\nBruh. Just bruh.";
170     }
171 }
172 JOptionPane.showMessageDialog(null, message, "Goodbye", JOptionPane.INFORMATION_MESSAGE);
173 }
174
175 /**
176  * Calculates the average of all grades.
177  *
178  * @return average of grades, or 0.0 if no grades
179  */
180 private static double calculateAverage() {
181     if (grades.isEmpty()) return 0.0;
182     double sum = 0.0;
183     for (double g : grades) sum += g;
184     return sum / grades.size();
185 }
```

The screenshot shows a Java IDE window with the file `GradeCalculator.java` open. The code defines a `getLetterGrade` method that takes a `double average` and returns a letter grade string based on the following scale:

Average Range	Letter Grade
≥ 93.0	A
≥ 90.0	A-
≥ 87.0	B+
≥ 83.0	B
≥ 80.0	B-
≥ 77.0	C+
≥ 73.0	C
≥ 70.0	C-
≥ 67.0	D+
≥ 63.0	D
≥ 60.0	D-
Below 60.0	F

```
186
187  /**
188   * Determines the Letter grade (with plus/minus) for a given average.
189   *
190   * @param average the numeric average
191   * @return Letter grade string
192   */
193  private static String getLetterGrade(double average) {
194      if (average >= 93.0) return "A";
195      if (average >= 90.0) return "A-";
196      if (average >= 87.0) return "B+";
197      if (average >= 83.0) return "B";
198      if (average >= 80.0) return "B-";
199      if (average >= 77.0) return "C+";
200      if (average >= 73.0) return "C";
201      if (average >= 70.0) return "C-";
202      if (average >= 67.0) return "D+";
203      if (average >= 63.0) return "D";
204      if (average >= 60.0) return "D-";
205      return "F";
206  }
207
208
209
```

The IDE interface includes a menu bar (File, Edit, Selection, View, ...), a search bar, and a status bar at the bottom showing "Ln 209, Col 1" and "Spaces: 4". A "Restricted Mode" warning is visible at the top.

Output:

















